

Multimodal Alpha Forecasting and Robust Portfolio Optimization

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1 Problem Summary

Our project aims to develop a multimodal machine learning and optimization framework to forecast short-term stock returns and then construct risk-aware portfolios. Using historical data for S&P 500 companies, we integrate technical indicators, fundamental ratios, and FinBERT-based sentiment from financial news and corporate filings. The model predicts next-day excess returns, which are then translated into optimal portfolio weights through CVaR optimization, linking predictive modeling with prescriptive decision-making.

2 Datasets

We construct a dataset combining market, fundamental, textual, and macro information for S&P 500 equities. The aim is to capture complementary signals from both quantitative trading activity and qualitative corporate disclosures. Table 1 in the Appendix lists the datasets and features that we propose to use.

3 Methods

Predictive Stage - Next-Day Return Forecasting

(1) Construct a panel of daily market indicators (returns, momentum, volatility), quarterly fundamentals (P/E, ROE, leverage, growth), and macro variables (VIX, yield spread). **(2)** Apply FinBERT to 10-K/10-Q MD&A sections and financial news to extract sentiment scores and text embeddings, merged with tabular data. **(3)** Train XGBoost, XGBoost + FinBERT embeddings, Optimal Regression Tree to predict the stocks' next day return.

Prescriptive Stage - Portfolio Optimization

Predicted returns feed into a mean-CVaR optimization that maximizes expected return and controls tail risk:

$$\max_w \mathbb{E}[r^\top w] - \lambda \text{CVaR}_\alpha(w),$$

Weights represent the portfolio decisions allocating capital across assets based on predicted returns.

4 Challenges

(1) Look-ahead bias: We will mitigate this by aligning fundamentals and filings strictly by their release dates and applying rolling-window validation for training and testing. **(2) High-dimensional text features:** FinBERT embeddings are computationally intensive; we will reduce dimensionality using PCA and apply regularization to prevent overfitting. **(3) Project scope:** Given time constraint, we may prioritize the predictive modeling stage if portfolio optimization cannot be fully implemented.

A Dataset and Feature Summary

Table 1: Summary of datasets and key features.

Dataset	Source	Frequency	Key Features / Variables
Market	Yahoo Finance, WRDS (CRSP)	Daily	Open, High, Low, Close, Volume (OHLCV); daily return; 5D/20D momentum; 10D/20D realized volatility; MA5/MA20 ratio; RSI; turnover (Volume / Shares Outstanding).
Fundamentals	WRDS (Compustat), AlphaVantage API	Quarterly → Daily	P/E, P/B, ROE, ROA, gross margin, Debt/Assets, EPS and revenue YoY growth, log(Market Cap). Forward-filled daily after each filing.
Textual (News)	NewsAPI, GDELT, FinBERT	Daily	FinBERT sentiment mean and volatility (7D, 30D); sentiment momentum ($\Delta 7D$); positive/neutral/negative sentiment ratios; news volume (headline count).
Textual (Filings)	SEC EDGAR, FinBERT	Quarterly → Daily	FinBERT sentiment of MD&A and Risk Factors; cosine similarity across filings (tone shift); sentiment change vs previous quarter; PCA-reduced FinBERT embeddings.
Macro	FRED, Nasdaq	Daily	VIX, Treasury yield spread (10Y–2Y), PMI, Federal Funds Rate, and S&P 500 index returns.